Compilation and parallelization gains generalize across formats

Speedup over the fully-interpreted baseline, geomean of regime geomeans over the 10-query workload · bid_strings, 31.3M rows

output format

input	CSV	JSON	HL7	XML
CSV base 5.37 Mrows/s	**96×†** (6.1× | 15.8×)	**67×** (5.5× | 12.2×)	**58×** (5.3× | 11.0×)	**65×** (5.5× | 12.0×)
JSON base 4.76 Mrows/s	**61×†** (3.3× | 18.3×)	**55×** (3.4× | 16.2×)	**50×** (3.4× | 14.9×)	**55×** (3.4× | 16.1×)
HL7 base 1.68 Mrows/s	**170×†** (10.3× | 16.6×)	**149×** (10.0× | 14.9×)	**138×** (10.3× | 13.4×)	**151×** (10.3× | 14.6×)
XML base 1.29 Mrows/s	**224×†** (13.2× | 17.0×)	**193×** (12.7× | 15.2×)	**178×** (12.9× | 13.9×)	**197×** (13.2× | 14.9×)

total

comp. | par.

gain over interpreted baseline
(rungs cumulative: total = comp. × par.)
× from compilation (JIT, W1)
× from parallelization (23 workers)

† memory-wall-clipped lower bound: ≥ half the workload runs ≥ 85% of the ~66 GB/s aggregate ceiling at 23 workers (filter-class queries touch the wall in every cell). in-memory sources isolate processing from I/O — the source is never the binder. baselines hand-implement identical codecs; interpreted rung materializes full records.